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SESSION 5

Task 1: Single Quotes and Literal Content

Task:

Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, and Test Edge Cases.

```
root@HP:~# nano session5_quotes.c
root@HP:~# gcc session5_quotes.c -o session5_quotes
root@HP:~# ./session5_quotes
Quote Parser Demo
=====
Enter a command containing single or double quotes.
Type exit to quit.

myshell> echo 'hello world'

Input: echo 'hello world'
Parsed content:
echo hello world
Quote parsing successful.

myshell> |

myshell> echo "hello world"

Input: echo "hello world"
Parsed content:
echo hello world
Quote parsing successful.

myshell> |
```

Task 2: Double Quotes and Variable Expansion

Task:

Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, and Test Quoted Commands.

```
myshell> echo "hello world"
```

```
Input: echo "hello world"
```

```
Parsed content:
```

```
echo hello world
```

```
Quote parsing successful.
```

```
myshell> echo 'Operating Systems Shell Programming'
```

```
Input: echo 'Operating Systems Shell Programming'
```

```
Parsed content:
```

```
echo Operating Systems Shell Programming
```

```
Quote parsing successful.
```

```
myshell> |
```

```
Quote parsing successful.
```

```
myshell> echo 'hello world
```

```
Input: echo 'hello world
```

```
Parsed content:
```

```
echo hello world
```

```
Error: Unclosed quote detected.
```

```
myshell> |
```

```
myshell> echo 'hello' "world"

Input: echo 'hello' "world"
Parsed content:
echo hello world
Quote parsing successful.

myshell> |

myshell> export NAME="Neeharika"

Input: export NAME="Neeharika"
Parsed content:
export NAME=Neeharika
Quote parsing successful.

myshell> echo $NAME

Input: echo $NAME
Parsed content:
echo $NAME
Quote parsing successful.

myshell> echo '$NAME'

Input: echo '$NAME'
Parsed content:
echo $NAME
Quote parsing successful.

myshell> echo "$NAME"

Input: echo "$NAME"
Parsed content:
echo $NAME
Quote parsing successful.

myshell>
```

Overall Observation – Session 5

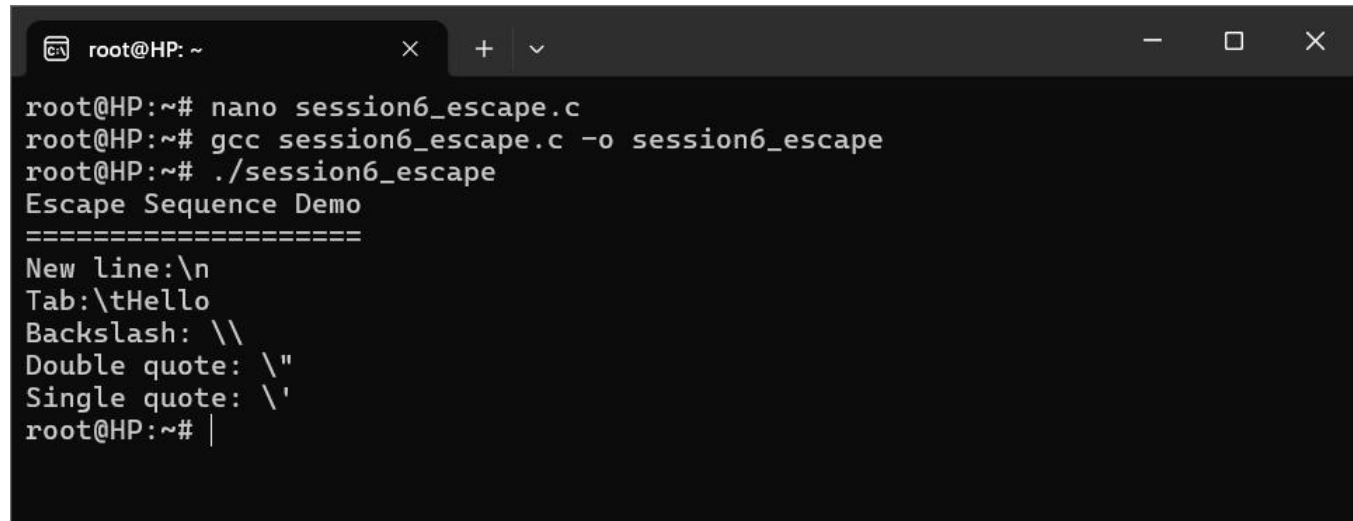
Session 5 focused on handling quoted input within the shell parser. Single quotes were used to preserve literal content without variable expansion, while double quotes allowed spaces to be preserved and variables to be expanded. Testing different quoted commands and edge cases provided practical understanding of how quoting affects command parsing and interpretation.

SESSION 6

Task 1: Escape Sequences and Special Character Handling

Task:

Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, and Test Complex Inputs.



```
root@HP: ~  
root@HP:~# nano session6_escape.c  
root@HP:~# gcc session6_escape.c -o session6_escape  
root@HP:~# ./session6_escape  
Escape Sequence Demo  
=====
```

The terminal window shows the execution of a C program named `session6_escape.c`. The program outputs the text "Escape Sequence Demo" followed by a line of equals signs. Below this, it lists several escape sequences and their corresponding characters: "New line: \n", "Tab: \tHello", "Backslash: \\", "Double quote: \\"", "Single quote: \'", and "root@HP:~# |".

Task 2: Child Process Creation and Command Execution

Task:

Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, and Test Command Launching.

```

root@HP:~# nano session6_execute.c
root@HP:~# gcc session6_execute.c -o session6_execute
root@HP:~# ./session6_execute
Starting command execution...
Parent process: waiting for child...
Child process: executing ls -l
total 320
-rw-r--r-- 1 root root    0 Aug  2 05:29 demo1.c
-rwxr-xr-x 1 root root 16088 Aug  2 05:44 demo2
-rw-r--r-- 1 root root   516 Aug  2 05:44 demo2.c
-rwxr-xr-x 1 root root 16280 Aug  1 04:16 fork_example
-rw-r--r-- 1 root root   591 Aug  1 04:15 fork_example.c
-rw----- 1 root root    27 Aug  1 03:52 nano.867.save
drwxr-xr-x 2 root root  4096 Jul 31 18:23 neeha
drwxr-xr-x 2 root root  4096 Jul 31 18:23 neetu
-rwxr-xr-x 1 root root 16136 Aug  8 03:18 process
-rw-r--r-- 1 root root   706 Aug  8 03:16 process.c
-rwxr-xr-x 1 root root 16128 Aug 20 09:59 session1
-rw-r--r-- 1 root root   441 Aug 20 09:58 session1.c
-rwxr-xr-x 1 root root 16016 Aug 20 10:13 session1_exec
-rw-r--r-- 1 root root   215 Aug 20 10:13 session1_exec.c
-rwxr-xr-x 1 root root 16184 Aug 20 10:16 session2
-rw-r--r-- 1 root root   349 Aug 20 10:10 session2.c
-rwxr-xr-x 1 root root 16200 Aug 20 10:19 session2_keyboard
-rw-r--r-- 1 root root   627 Aug 20 10:19 session2_keyboard.c
-rwxr-xr-x 1 root root 16320 Aug 20 14:13 session3_history
-rw-r--r-- 1 root root  1227 Aug 20 14:13 session3_history.c
-rwxr-xr-x 1 root root 16184 Aug 20 14:15 session3_memory
-rw-r--r-- 1 root root  1030 Aug 20 14:15 session3_memory.c
-rwxr-xr-x 1 root root 16232 Aug 20 14:25 session4_parser
-rw-r--r-- 1 root root  1193 Aug 20 14:25 session4_parser.c
-rwxr-xr-x 1 root root 16200 Aug 20 14:23 session4_tokenizer
-rw-r--r-- 1 root root   611 Aug 20 14:23 session4_tokenizer.c
-rwxr-xr-x 1 root root 16280 Aug 20 14:27 session5_quotes
-rw-r--r-- 1 root root  1258 Aug 20 14:27 session5_quotes.c
-rwxr-xr-x 1 root root 15968 Aug 20 14:34 session6_escape
-rw-r--r-- 1 root root   291 Aug 20 14:34 session6_escape.c
-rwxr-xr-x 1 root root 16144 Aug 20 14:35 session6_execute
-rw-r--r-- 1 root root   545 Aug 20 14:35 session6_execute.c
drwxr-xr-x 3 root root  4096 Aug 20 09:57 shellforge
drwxr-xr-x 2 root root  4096 Jul 31 18:16 sri
-rwxr-xr-x 1 root root 16136 Aug  8 03:18 states
-rw-r--r-- 1 root root   576 Aug  8 03:17 states.c
-rw-r--r-- 1 root root    0 Jul 31 18:28 xls.txt
Parent process: child finished.
root@HP:~# |

```

```

root@HP:~# nano session6_exec_date.c
root@HP:~# gcc session6_exec_date.c -o session6_exec_date
root@HP:~# ./session6_exec_date
Before exec()
Thu Aug 20 14:37:07 UTC 2026
root@HP:~# |
root@HP:~# nano session6_error.c
root@HP:~# gcc session6_error.c -o session6_error
root@HP:~# ./session6_error
Attempting to execute a command...
exec failed: No such file or directory
root@HP:~# |

```

Overall Observation – Session 6

Session 6 focused on two important aspects of shell implementation: escape-sequence handling and process-based command execution. The first task demonstrated how escaped spaces and special characters can be preserved during parsing. The second task provided practical experience with creating child processes, executing programs, passing arguments, and managing the parent process. The activities helped demonstrate how parsed shell commands are ultimately prepared for execution.